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Effective Options Utilized to Wellbore Engineering Data Evaluation and Analysis for Drilling Performance Improvement

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Abstract

To mitigate drilling risks and enhance the rate of penetration (ROP) in shale gas horizontal wells, wellbore stability analysis and geomechanical studies are essential. However, cluster development of horizontal wells within the same platform alters the original in-situ stress field, rendering the drilling mud weight window calculated based on static mechanical models inaccurate. This frequently leads to wellbore instability during drilling, posing risks to both drilling safety and environmental protection. This paper integrates actual drilling, fracturing, and production data to establish a multidimensional geomechanical model. Building upon a 1D static geomechanical model, it incorporates stress field variations in the target drilling area caused by fracturing and production operations from three previously completed horizontal wells in the northern cluster. Key findings include: 1. Dynamic Stress Field Analysis: After injecting approximately 100,000 m³ of fracturing fluid into three northern horizontal wells and considering cumulative production effects, the "fracturing-production" operations in the northern cluster induced a pore pressure increase of 0.6–1.6 MPa in the southern platform area. This corresponds to a mud weight increment of 0.07–0.1 g/cm³, necessitating adjustments to drilling fluid density for enhanced wellbore stability. 2. Target Formation Mechanical Properties: Poisson's ratio: 0.152–0.22. Young's modulus: 6.1–65.8 GPa. Rock strength: 60.7–117 MPa. The low Poisson's ratio and high Young's modulus indicate medium-hard formations with favorable drillability. 3. Formation Pressure Evolution: The Hanjiadian Formation exhibits gradual pressure buildup, with the Longmaxi source rock reaching a maximum pressure gradient of 1.49 g/cm³. This dynamic geomechanical model demonstrates strong applicability and accuracy in predicting wellbore stability and optimizing mud weight windows. It provides critical insights for drilling design, effectively addressing stress field perturbations caused by multi-well interference in cluster development. For instance, the quantified mud weight adjustment (0.07–0.1 g/cm³) directly translates to safer drilling practices while avoiding excessive mud costs – a balance crucial for both engineering safety and economic efficiency in shale gas operations.

Introduction

Wellbore stability refers to a technical measure that maintains the structural stability of the wellbore during the well drilling and completion processes, preventing the collapse or deformation of the wellbore. The accurate calculation of stress in the wellbore area and the rock strength instability (failure) criterion are the

keys to studying wellbore stability. Kirsch was the first to use the method of elasticity mechanics and gave the elastic equation of stress concentration around the borehole under the assumption of plane strain. After that, Westergaard used the elastoplastic theory to obtain the analytical solution of the elastoplastic stress of the surrounding rock of the wellbore when there is no mud pressure in the wellbore and only under the uniform action of the original field horizontal stress. And the Mohr-Coulomb criterion was used to conduct a simple stability analysis of the surrounding rock of the wellbore. Biot considered that microcracks would occur in the surrounding rock of the wellbore under the action of stress and proposed the classical theory of porous elastic media, which was quickly applied to wellbore stability analysis. Bradley studied the elastic solution of the surrounding rock of the wellbore considering pore pressure under the condition of plane strain. Assuming that the formation is homogeneous and isotropic, and there is no interaction between the drilling fluid and the fluid in the pores, it was considered that the tangential stress represents the main stress concentration effect around the wellbore. Therefore, the determination of the mechanical analysis model and research method of wellbore stability is not only affected by the complexity of the rock itself but also by the experimental testing methods and technical limitations of its properties. Ci Jianfa et al. compiled a corresponding finite element calculation model based on the principle of rock fracture damage and the characteristics of shale hydration swelling, and obtained a calculation method for the safe mud density value required to maintain wellbore stability on the basis of the Mohr-Coulomb criterion. At present, the research on the stress analysis of surrounding rock at home and abroad shows that the linear elastic method is still the main method for studying wellbore stability problems. Aiming at the wellbore instability problem in the shale gas reservoir in southern Sichuan, this paper proposes a method for determining the value of the safe mud density window based on considering the in-situ stress changes near the wellbore caused by the fracturing and production of adjacent wells on the same platform, combined with the equivalent mud density of pore pressure and the minimum horizontal principal stress. The application results of actual data show that the proposed method has good applicability and accuracy and can provide an effective reference for construction design.

Theoretical Basis of Wellbore Stability Control Technology

Instantaneous Elastic Analysis Method

Assuming the formation is a homogeneous, isotropic, linear elastic material, and the wellbore surrounding rock is in a plane strain state, while neglecting the physicochemical interactions between the rock and drilling fluid as well as thermal stress effects. The instantaneous elastic analysis method can be used to calculate the stress distribution around the wellbore under varying stress conditions, thereby determining the collapse pressure and fracture pressure. Collapse pressure: The minimum pressure at which the wellbore rock collapses under compressive stress. Fracture pressure: The minimum pressure at which the wellbore rock fractures under tensile stress. Based on the calculated results, the required drilling fluid density can be determined to ensure wellbore stability. This method effectively predicts and controls wellbore instability risks, enhancing drilling safety and efficiency.

Finite Element Analysis Method

In the study of wellbore stability, the finite element analysis method can simulate the deformation and failure processes of the wellbore under the influence of in-situ stress and drilling fluid pressure. By establishing a finite element model of the wellbore and its surrounding rock layers, and inputting the mechanical parameters and initial conditions of the formation, it is possible to analyze in detail the stress distribution, deformation characteristics, and creep behavior of the wellbore under different drilling conditions. This method not only considers the anisotropic and nonlinear properties of the formation but also can simulate the long-term creep and instantaneous elastic deformation of the wellbore under different geological conditions. Through finite element analysis, the density and pressure control scheme of the drilling fluid can be

optimized to prevent wellbore instability accidents, providing important technical support and theoretical basis for drilling operations under complex geological conditions.

Analysis and Calculation of Wellbore Stress State

Using the mechanical parameters of the formation (such as elastic modulus, Poisson's ratio, etc.) and the distribution of the in-situ stress field around the wellbore, the triaxial stress distribution of the wellbore can be determined through stress analysis methods. Assuming that the formation is a homogeneous isotropic material and the surrounding rock of the wellbore is in a plane strain state, the stress state at a certain point in the wellbore can be represented as:

$$\sigma_{rr} = p_m \sigma_{\theta\theta} = \frac{\sigma_{zz} + \sigma_{\theta\theta}}{2} - \sqrt{\left(\frac{\sigma_{zz} - \sigma_{\theta\theta}}{2}\right)^2 + \tau_{\theta z}^2} \quad \sigma_{zz} = \frac{\sigma_{zz} + \sigma_{\theta\theta}}{2} + \sqrt{\left(\frac{\sigma_{zz} - \sigma_{\theta\theta}}{2}\right)^2 + \tau_{\theta z}^2} \quad (1)$$

σ_{rr} and σ_{zz} are the radial stress, circumferential stress, and axial stress, respectively, in Pa; p_m is the mud column pressure, in Pa; and $\tau_{\theta z}$ is the shear stress, in Pa. To calculate the stress state of the wellbore, it is first necessary to obtain parameters such as the elastic modulus and Poisson's ratio of the formation. These parameters are typically acquired through laboratory tests or geological surveys. Then, using these parameters and the distribution of the in-situ stress field, the triaxial stress distribution of the wellbore is calculated through stress analysis methods. The specific steps include determining the initial stress state around the wellbore, applying stress equilibrium equations and material constitutive relationships, and calculating the stress distribution of the wellbore at different depths and drilling conditions. Collapse pressure refers to the minimum pressure at which the wellbore rock collapses under compressive stress, while fracture pressure is the minimum pressure at which the wellbore rock fractures under tensile stress. According to Mohr's circle theory and the Coulomb-Mohr failure criterion, the collapse pressure can be calculated using the following formula:

$$\sigma_1 = \sigma_3 \tan^2\left(45^\circ - \frac{\phi}{2}\right) + 2C \tan\left(45^\circ - \frac{\phi}{2}\right) \quad (2)$$

σ_1 and σ_3 are the maximum and minimum principal stresses, respectively, in Pa; ϕ is the internal friction angle, in degrees; and C is the cohesion of the rock, in Pa. In practical calculations, the actual mechanical parameters of the surrounding rock of the wellbore are selected, and combined with instantaneous elasticity theory, the collapse and fracture pressures at different depths are calculated.

Geomechanical Modeling

Based on conventional logging data and imaging data, a reservoir geomechanical model is established. The model parameters are calibrated using data from drilling, formation testing, fracturing tests, and core experiments (Figure 1). The research results are used to guide the mud weight design for the drilling engineering plan of the Y102H36 platform.

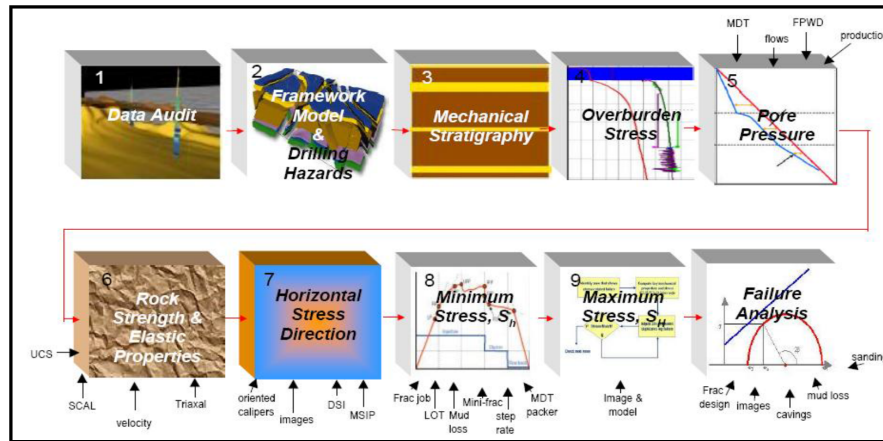


Figure 1—Geomechanical Model Workflow

Rock Mechanics Modeling

Dynamic elastic modulus is calculated using the P-wave and density logs. The elastic modulus is referred to as dynamic because the acoustic logging is conducted at high frequencies (approximately 10 kHz). Compared to mechanical property testing in laboratories or under stable loads at the surface/near the wellbore, rocks exhibit stiffer characteristics under dynamic conditions. The dynamic elastic modulus and Poisson's ratio are calculated using the following correlation:

$$E_{\text{dyn}} = \rho V_s^2 (3V_p^2 - 4V_s^2) / (V_p^2 - V_s^2) \quad (3)$$

$$\nu_{\text{dyn}} = (V_p^2 - 2V_s^2) / 2(V_p^2 - V_s^2) \quad (4)$$

Elastic Modulus. The elastic modulus E of a rock refers to the ratio of axial stress to the increment of axial strain under uniaxial compression conditions, as shown in the following equation:

$$E = \Delta \sigma_z / \Delta \varepsilon_z \quad (5)$$

In this equation, E —the elastic modulus—is the slope of the stress-strain curve, representing the rate of change of stress with respect to strain under uniaxial stress conditions.

$\Delta \sigma$, $\Delta \varepsilon_z$ —increments of axial stress and strain.

It has a significant impact on fracture width and fracturing pressure. In a two-dimensional model where the fracture height is assumed to be constant, for Newtonian fluids, the fracture width is inversely proportional to the fourth root of the elastic modulus, i.e., $W \sim 1/E^{1/4}$; for non-Newtonian fluids, the relationship between fracture width and elastic modulus is $W \sim 1/E^{1/(2n'+2)}$, where n' is the power-law index of the non-Newtonian fluid. Whether in a two-dimensional model or a three-dimensional model, calculations show that as the elastic modulus of the rock increases, the fracture width decreases.

According to the two-dimensional fracture model, for Newtonian fluids, the net pressure during fracturing operations is proportional to the 3/4 power of the elastic modulus, i.e., $\Delta p(t) \sim E^{3/4}$; for non-Newtonian fluids, the relationship between pressure and elastic modulus is $\Delta p(t) \sim E^{(2n'+1)/(2n'+2)}$. As the elastic modulus increases, the net pressure during the operation will also increase.

Poisson's ratio. The Poisson's ratio of a rock refers to the ratio of lateral strain to axial strain under uniaxial compression conditions.

$$\nu = -\frac{\varepsilon_r}{\varepsilon_z} \quad (6)$$

Poisson's ratio is an important parameter in determining horizontal ground stress. Generally, due to the relationship between lateral strain and axial strain, as well as the relationship between axial strain and axial

stress, which are often in the form of curves, the Poisson's ratio of rock tends to increase with increasing axial stress. The linear segment of the stress-strain curve is commonly used as the basis for calculations.

The deformation modulus and Poisson's ratio of a rock mass are influenced by various factors, including the mineral composition, structure and texture of the rock, degree of weathering, porosity, moisture content, microstructural surfaces, and their relationship with the load direction, leading to significant variability.

Shear Modulus. The shear modulus is the ratio of shear stress to strain within the elastic deformation limit under shear stress, also known as the modulus of rigidity or rigidity modulus. It is one of the mechanical property indicators of a material, representing the material's ability to resist shear strain. A larger modulus indicates stronger rigidity, making the rock more resistant to shear deformation; when the shear modulus of a rock is infinite, the rock becomes a rigid body. The reciprocal of the shear modulus is called the shear compliance, which measures the amount of shear strain under a unit shear force, indicating the ease of shear deformation of the material.

The shear modulus G of a rock is related to its elastic modulus E and Poisson's ratio ν as follows:

$$G = \frac{E}{2(1+\nu)} \quad (7)$$

Rock Compressive Strength. This refers to the ultimate strength of a rock specimen under uniaxial pressure at the point of failure, numerically equal to the maximum axial stress at failure, usually denoted as σ_c :

$$\sigma_c = \frac{P}{A} \quad (8)$$

In the equation, P represents the applied load at failure, known as the failure load; A is the original cross-sectional area.

If the results from rock mechanics laboratory measurements are available, this correlation will be calibrated by matching the rock mechanical properties derived from well logging with the laboratory measurement results.

In this context, UCS is the uniaxial compressive strength, Δt is the sonic transit time with a unit of $\mu s/ft$, $NPHI$ is the porosity with a unit of 1, and E is the Young's modulus with a unit of Pa. By calibrating the computational model using the triaxial compression test results from cores in the Sun region, the results show that the Poisson's ratio of the target Longmaxi Formation in the Yang 102H36-1 well group ranges from 0.152 to 0.227, the Young's modulus ranges from 36.1 to 65.8 GPa, and the rock strength ranges from 60.7 to 117 MPa (Table 1). This indicates characteristics of low Poisson's ratio and high Young's modulus, classifying the formation as a medium-hard formation with relatively good drillability (Figure 2).

Table 1—Statistical Results of Rock Mechanics Parameters for Well Group Y102H36

Formation Name	Young's Modulus (GPa)	Poisson's Ratio (Dimensionless)	UCS (MPa)
Jialingjiang Formation	\	\	\
Feixianguan Formation	19.7~47.5	0.182~0.242	28.4~115.2
Leping Formation Coal Seam Section	3.6~8.3	0.362~0.421	7.8~21.7
Leping Formation Non- Coal Seam Section	23.6~44.2	0.195~0.231	56.5~89.5
Maokou Formation	38.4~57.2	0.147~0.267	67.7~158.3
Qixia Formation	40.5~55.0	0.153~0.257	59.3~155.1
Liangshan Formation	38.3~63.6	0.135~0.275	96.6~107.8
Hanjiaodian Formation	27.6~55.3	0.137~0.221	49.2~129.8
Shiniulan Formation	26.8~63.3	0.129~0.245	44.5~111.3
Longmaxi Formation	36.1~65.8	0.152~0.227	61.7~119.9

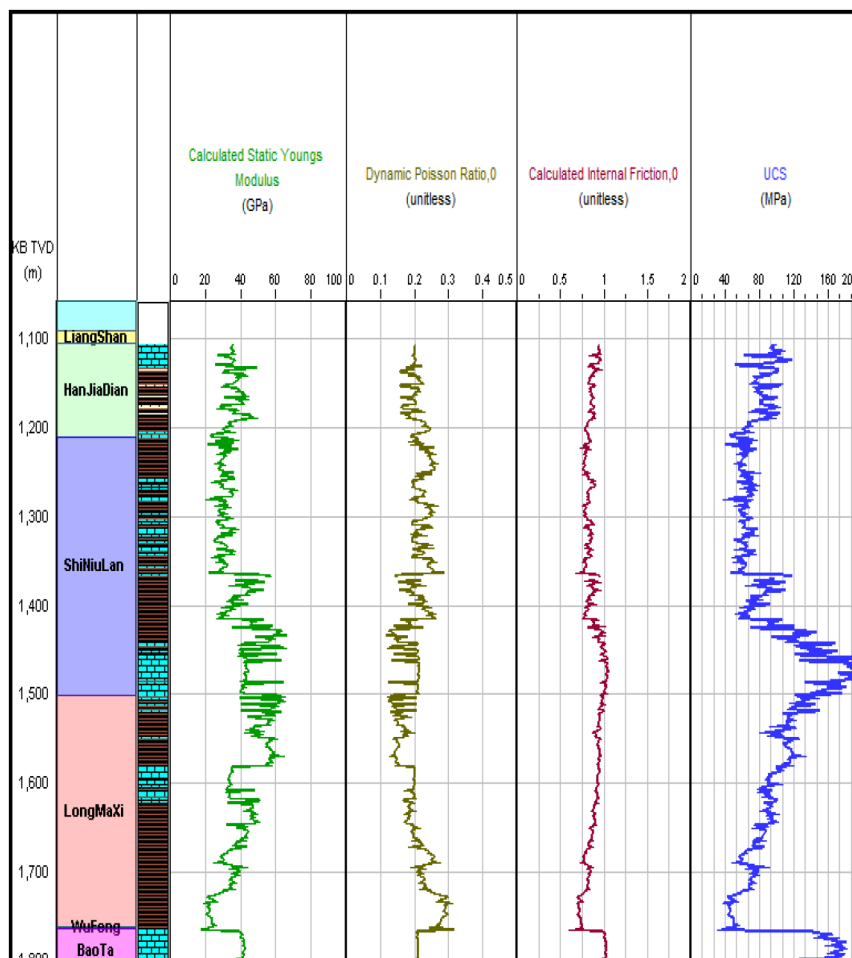


Figure 2—Rock Mechanics Modeling Results of Drilled Wells

Pore Pressure Analysis

The undercompaction method is based on the trend of effective stress and P-wave velocity. This technique defines the "normal" evolution of porosity, whereby P-wave velocity increases with increasing effective stress, i.e., with vertical depth. Overpressured formations will be in an undercompacted state because the pores remain open under high pressure. The acoustic velocity in undercompacted formations will be slower than the expected velocity for that formation depth, as predicted by the trend line (the blue part in the figure). Any difference between the expected acoustic velocity (trend line) and the actual/measured velocity is associated with overpressure. Calibrating the compaction trend on offset wells is a critical step in the analysis. The first step is to establish the compaction trend on the offset well. Compaction is significantly influenced by lithology, and to maintain consistency, the trend line applies only to shale (Figure 3).

Through the above undercompaction method analysis, combined with the actual measured pore pressure from offset wells and gas influx and kick events during the drilling process, calibration is performed. Then, the calibrated trend line is used to predict the pore pressure for the relevant horizontal wells at the Yang 102H36 platform.

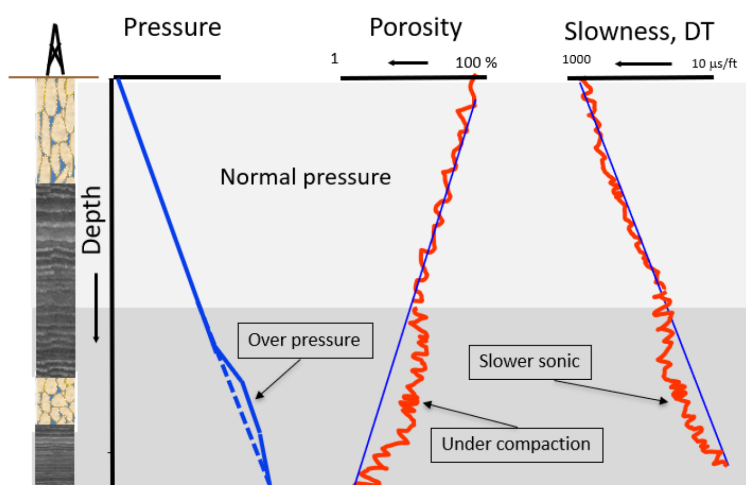


Figure 3—Schematic Diagram of Pore Pressure Prediction Method Based on Undercompaction

Formation pore pressure is an important aspect of geomechanical modeling and holds significant meaning in the context of regional stress variations. Pore pressure directly affects drilling, completion, and oil and gas production operations throughout the entire lifecycle of a reservoir. The typical workflow for one-dimensional pore pressure modeling starts with estimating post-drilling pore pressure using data from nearby wells, especially data that directly monitors formation pressure. This modeling involves detailed preprocessing and environmental correction of all logging data from nearby wells. This process requires consideration of all geological-related data, including logging data, formation pressure test data, and drilling events, along with necessary quality control and logging data analysis. In addition, different geological features and environments have various influencing factors on the generation of formation overpressure. Understanding the mechanisms of overpressure and the impact of different geological parameters on overpressure is crucial not only for pore pressure prediction but also for the overall accuracy of the geomechanical model.

Due to the presence of several sets of thick hydrocarbon source rock layers, along with the aging of formations and a complex tectonic history, the overpressure mechanisms in the study area are relatively complex. They may primarily consist of secondary pressure generation mechanisms caused by factors such as fluid expansion and hydrocarbon generation. Methods for predicting pore pressure based on logging data, such as the normal compaction trend line method, cannot accurately predict overpressure. However, the characteristics of overpressure in shale formations still show good responses on logging curves. By conducting a comprehensive analysis of drilling history, regional pressure data, geological information, and calibrating the geomechanical model, model parameters can be adjusted, and logging curves can be used to assist in pore pressure prediction. Based on regional well testing data and drilling history analysis, both the overlying formations and the target formations exhibit high pressure, and no depletion has been reported in the study area. Due to regional differences in burial depth, organic carbon content, hydrocarbon generation, and low permeability characteristics, there is significant variation in pore pressure within the target formations.

Predicting the pore pressure for the target well: The formations above and including the Maokou Formation exhibit a hydrostatic pressure gradient. However, the presence of thin coal seams in the Leping Formation and natural fractures in parts of the Maokou Formation may lead to certain gas measurement anomalies. Pressure begins to gradually increase from the base of the Qixia Formation, reaching approximately 1.30 g/cm^3 at the bottom of the Shiniulan Formation. After entering the Longmaxi Formation, the pressure coefficient continues to increase. In the horizontal section of this well, the original formation pressure coefficient is 1.49 g/cm^3 , which does not account for the fracturing and production conditions of the northern half branch.

4D Geomechanics Analysis

During the development of the Yang 102H36 platform, the injection of large volumes of fracturing fluid into the formation caused changes in the reservoir stress field. To simulate the stress field changes during fracturing operations and production development, four-dimensional stress research requires multi-field coupling to establish the pattern of changes in the reservoir stress model over time. Conducting multidimensional deep reservoir multi-field coupled dynamic geomechanical simulations characterizes the stress field's variation over time during development. As formation pressure continuously changes during oil and gas development, the geostress will also change accordingly, and the stress state within the reservoir is in a constant state of flux. Using techniques such as reservoir numerical simulation and geostress simulation, this study examines the changes in pore pressure within the reservoir during development. The research results will thus provide drilling fluid density design for the four horizontal wells yet to be drilled in the southern half of the Yang 102H36 platform.

In the long-term injection and production process of reservoirs, the numerical simulation method for the dynamic evolution of geomechanics involves using a hydraulic fracturing fracture propagation model. Hydraulic fracture parameters are simulated and calculated based on fracturing operation data. Initial pore pressure and geomechanics are used as the initial conditions for the geomechanical model. The pore pressure change data obtained from the seepage model during the reservoir injection and production process serves as the boundary conditions for conducting seepage-geomechanical coupled simulations during reservoir injection and production. This method takes into account the impact of stress changes on the formation's porosity and permeability parameters, incorporating a stress sensitivity calculation model. Iterative coupling meets the requirements for formation heterogeneity, while maintaining high computational efficiency and accuracy.

Based on the established geological model and the fracture network expansion model, the simulation of artificial fracture propagation in the fracturing scheme utilizes unstructured grid technology for numerical model preprocessing. Fine grids are employed around the horizontal well to accurately preserve the shape of artificial fractures, while coarser grids are used for other matrix parts, maximizing the accuracy of the numerical model.

The coarsened models of wells Yang 102H36-1, Yang 102H36-2, and Yang 102H36-3, including original saturation, permeability, porosity, and pressure fields, are imported into numerical simulation software to achieve detailed characterization of fracture morphology. This approach significantly improves the accuracy of numerical simulations compared to fracture equivalent simulation techniques (Figure 4).

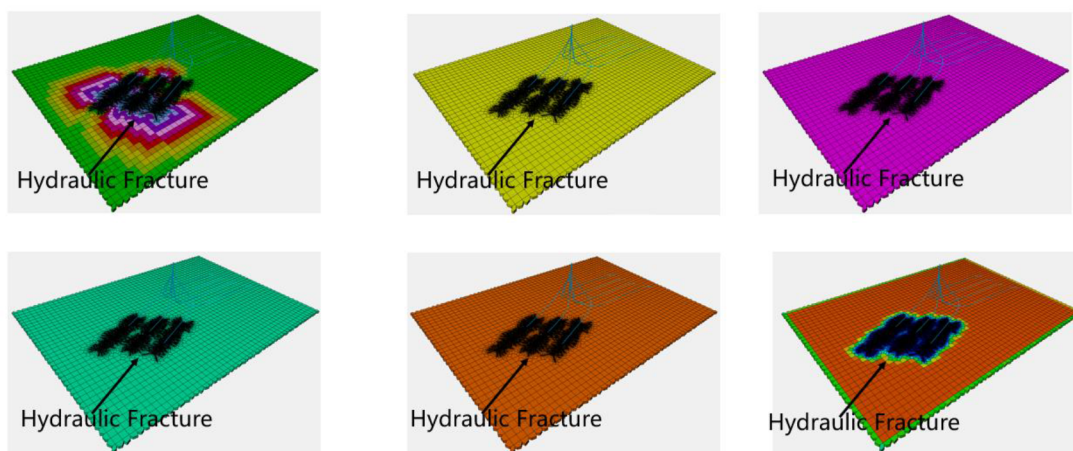


Figure 4—Full-Coupling Preprocessing of Y102H36 Platform Model

Based on the three-dimensional static geostress, a four-dimensional dynamic geostress study for a single well is conducted. During the entire four-dimensional stress simulation process, stress changes are divided into two phases: an increase in formation pressure during fracturing (pressurization phase) and a decrease in formation pressure during development (depressurization phase).

The changes in the production pore pressure field for wells Yang 102H36-1, Yang 102H36-2, and Yang 102H36-3 on the Yang 102H36 platform characterize the stress variation features at different production times, providing technical support for the areas yet to be drilled. As of now, the stress increment in the northern half of the platform ranges from 0.6 to 1.6 MPa. Considering various causes of abnormal high pressure and integrating the analysis results of pore pressure characteristics in the area where the Yang 102H36 well group is located and neighboring wells, along with the fracturing and production conditions of the three horizontal wells on this platform, the pore pressure for the target wells on the yet-to-be-drilled platform is dynamically predicted. Simulation results indicate that fracturing and production do not alter the pressure in the Shiniulan and overlying formations, but only affect the Longmaxi formation. After fracturing and production, the pore pressure coefficient in the horizontal section of the wells to be drilled increases to 1.60 g/cm³ (Figure 5).

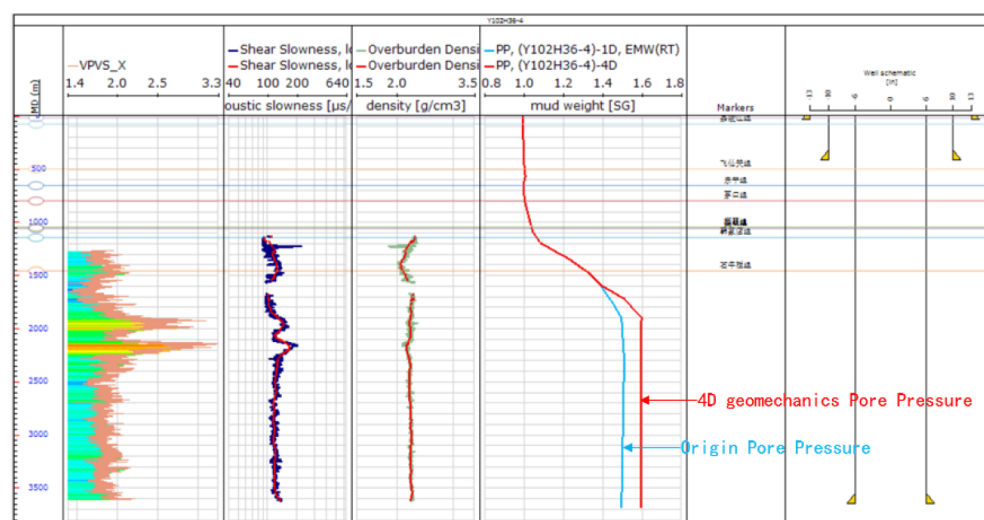


Figure 5—Pore Pressure Prediction for Well Yang 102H36-4

Statistics of Drilling Events in Adjacent Wells

In the Longmaxi Formation, frequent occurrences of wellbore leakage have been observed, with the equivalent mud density during these losses mainly ranging from 1.60 to 1.82 g/cm³. This is primarily characterized by continuous losses/seepage during drilling (drilling while simultaneously controlling losses) or encountering small faults (zero weight on bit, significant increase in hook load, and loss of returns at the wellhead). In the Qixia Formation, the equivalent mud density during losses ranges from 1.10 to 1.46 g/cm³. Occasional losses have also been reported in the Jialingjiang, Maokou, and Hanjiaodian Formations (Table 2).

Table 2—Regional Loss Data Statistics

Well Number	Formation	Depth	Mud Density at the Time of Loss
		m	g/cm ³
Yang 102H33-1	Jialingjiang Formation	95	1.00
	Qixia Formation	1300	1.46
	Longmaxi Formation	1906	1.80
		2217	1.80
		2665	1.80
		3068	1.75
Yang 102H34-2	Qixia Formation	950	1.15
	Hanjiaodian Formation	1066	1.15
		1106	1.15
	Longmaxi Formation	3151	1.66
Yang 102H34-3	Qixia Formation	865	1.17
		930	1.18
		942	1.10
	Longmaxi Formation	2931	1.66
		3058	1.60
Yang 102H34-4	Longmaxi Formation	2635	1.66
		2844	1.66
		2922	1.66
Yang 102H36-1	Longmaxi Formation	1987	1.82
Yang 102H36-2	Maokou Formation	906	1.30
YS145H	Longmaxi Formation	3572	1.65

After the uplift of the formation, due to stress release, weathering and erosion, and weak post-depositional compaction, the rocks above the Jialingjiang Formation exhibit weak cementation, making them loose and friable. During drilling, losses are inevitable, and frequent downhole issues such as obstructions, sticking, blockages, and sand settling can occur. Analysis of the mechanical and ground-breaking test results from the Yang region indicates that the minimum horizontal principal stress for the Qixia to Hanjiaodian Formations is approximately between 1.97 and 2.11 g/cm³, and for the Longmaxi Formation, it is approximately between 2.15 and 2.35 g/cm³. These values are higher than the mud density during losses, suggesting that the losses are related to the development of fractures and faults.

In the Longmaxi Formation, the frequency of sticking and obstruction during tripping operations is the highest, but the conditions vary across different wells, indicating strong heterogeneity in the formation. Generally, sticking incidents occur when using mud densities between 1.65 and 1.89 g/cm³; in some cases, sticking and losses occur in the same layer, posing significant challenges for drilling operations. In the Jialingjiang Formation, obstructions were encountered twice while running casing. In the Qixia Formation, sticking occurred once ([Table 3](#)).

Table 3—Regional Sticking Data Statistics

Well Number	Formation	Depth of Sticking	Mud Density at the Time of Sticking	Remarks
		m	g/cm ³	
Yang 102H32-1	Jialingjiang Formation	42	1.00	Obstruction encountered while running casing
		2970	1.75	Obstruction encountered while tripping in, with losses
Yang 102H33-1	Jialingjiang Formation	49	1.00	Obstruction encountered while running casing
	Longmaxi Formation	2970	1.75	Obstruction encountered while tripping in
Yang 102H34-3	Qixia Formation	928	1.10	Obstruction encountered at 928m, difficult to ream, sticking occurs every 10cm while pulling out the drill
Yang 102H34-4	Longmaxi Formation	3250	1.66	Obstruction encountered while tripping in, reaming successful
Yang 102H36-1	Longmaxi Formation	1987	1.82	Sticking encountered while tripping out
		2046	1.82	Indications while both pulling and lowering
		2159	1.85	Obstruction encountered while tripping in
		2215	1.85	Obstruction encountered while tripping in
		2230	1.85	Sticking encountered while tripping out
		2932	1.85	Indications while both pulling and lowering
		2034	1.85	Sticking encountered while tripping out
		2000	1.85	Obstruction encountered while tripping in
Yang 102H36-3	Longmaxi Formation	1612	1.71	Sticking encountered while pulling, obstruction while lowering
		2090	1.85	Sticking encountered while pulling, cuttings observed during circulation
		2618	1.86	Sticking encountered while tripping out
		3180	1.89	Obstruction encountered while tripping in
YS145H	Longmaxi Formation	2520	1.65	Obstruction encountered while tripping in
		3479	1.65	Obstruction encountered while tripping in during cleaning, loss of returns at the wellhead
		3496	1.65	Obstruction encountered while tripping in during cleaning, with losses

Table 4—Pressure Prediction Results for Different Formations in Yang 102H36-4

Formation Name	Pore Pressure (Original)	Pore Pressure (Considering Production)	Collapse Pressure (Oil-Based)	Collapse Pressure (Water-Based)	Fracture Pressure
	g/cm ³	g/cm ³	g/cm ³	g/cm ³	g/cm ³
Jialingjiang	1.0~1.01	1.0~1.01	\	\	\
Feixianguan	1.01~1.02	1.01~1.02	0.32~0.89	0.40~0.97	1.85~1.92
Leping	1.02~1.03	1.02~1.03	0.53~1.25	0.61~1.33	1.92~2.01
Maokou	1.01~1.05	1.01~1.05	0.53~0.72	0.61~0.80	1.93~1.97
Qixia	1.01~1.05	1.01~1.05	0.62~0.9	0.70~0.98	1.97~2.01
Liangshan	1.01~1.0.8	1.01~1.0.8	0.55~1.12	0.63~1.20	2.01~2.03
Hanjiaodian	1.08~1.13	1.08~1.13	0.82~1.26	0.90~1.34	2.03~2.11
Shiniulan	1.13~1.32	1.13~1.32	0.86~1.23	0.94~1.31	2.1~2.15
Longmaxi	1.32~1.49	1.32~1.60	1.58~1.70	1.66~1.78	2.15~2.35

The rocks in the formations above the Jialingjiang Formation are affected by stress release, weathering and erosion, and lack of compaction after uplift, resulting in poor cementation. These rocks are loose and friable, making them highly susceptible to varying degrees of collapse and fragmentation.

During drilling in the Qixia Formation of the Yang 102H34-3 well, a loss of circulation occurred, leading to a loss of mud returns at the wellhead. Despite multiple attempts at continuous grouting, the fluid level did not recover. Subsequently, a directive was issued to change the drill bit and continue drilling forcefully. While tripping in to 928 meters, obstructions were encountered, and reaming was difficult with persistent sticking. The analysis identified two main reasons: 1. Due to the loss of circulation and returns, cuttings were not effectively removed from the wellbore, leading to the formation of a sand bridge as they settled, causing obstructions during tripping, and sticking when pulling out; 2. The loss of returns resulted in a reduced effective mud density in the wellbore, weakening support for the wellbore walls, which led to new collapses and fragmentation in weaker formations, increasing the amount of cuttings in the wellbore (Figure 6).

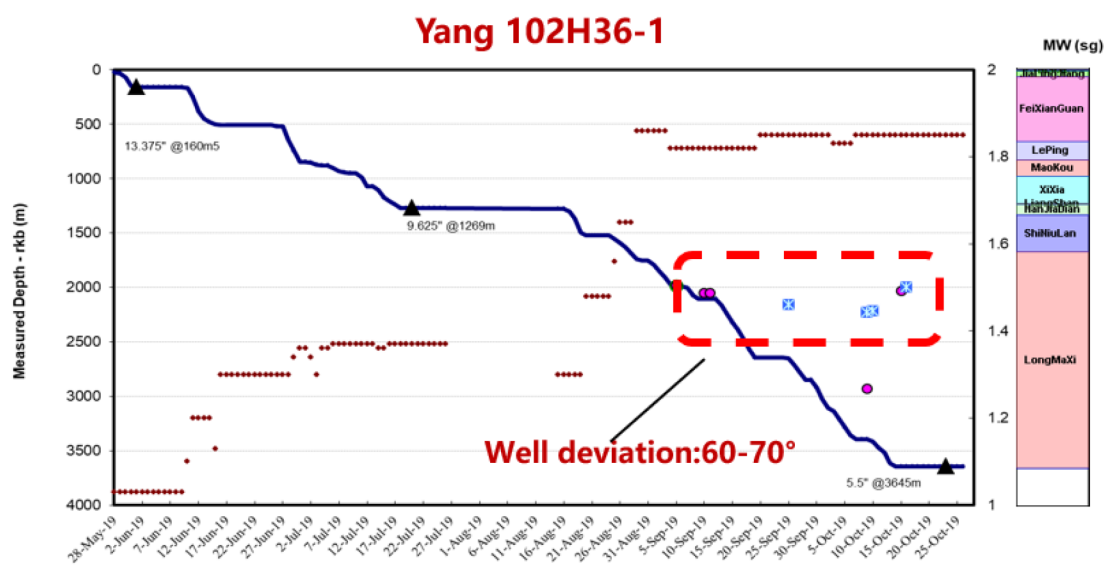


Figure 6—Sticking Analysis of Yang 102H36-1

In the Yang 102H36-1 well, sticking during tripping operations in the Longmaxi Formation mainly occurred in the well sections with an inclination angle between 60-70 degrees. Specifically, obstructions during tripping in were primarily encountered after drilling to 3300 meters. The caliper log indicates significant borehole enlargement, particularly beyond 2500 meters, where the enlargement is more severe than in the upper formations. From the collapse pressure perspective, the mud density in the Longmaxi Formation is generally lower than the required mud density. Due to the highly developed bedding planes and low-angle fractures in this area, it is determined that the sticking during tripping operations is related to the weak bedding planes and low-angle fracture surfaces.

Pre-Drilling Pressure Prediction Results

Since there are no core soaking experiments conducted in this area, we refer to research results from similar blocks in China. In these studies, cores were placed in containers filled with mud for soaking in laboratory experiments, and the mechanical parameters of the rocks, such as Young's modulus and Poisson's ratio, were recorded at different soaking times. Using the recorded mechanical parameter data points, formulas were derived to calculate the rock mechanical parameters under various soaking durations, providing dynamic values of these parameters as a function of soaking time. Experimental results show that with increased soaking time, rock strength gradually decreases, thereby increasing the risk of wellbore collapse (Table 3). Due to the differential effects of mud systems on rock strength during soaking, collapse pressure predictions were analyzed for different soaking times. The study results indicate that under the same soaking conditions, oil-based mud exhibits higher stability than water-based mud, demonstrating better inhibition. The average collapse pressure for oil-based mud is 0.08 g/cm³ lower than that of water-based mud, with the range of difference being related to rock strength. As soaking time increases, the difference between the two mud systems also increases (Figure 7).

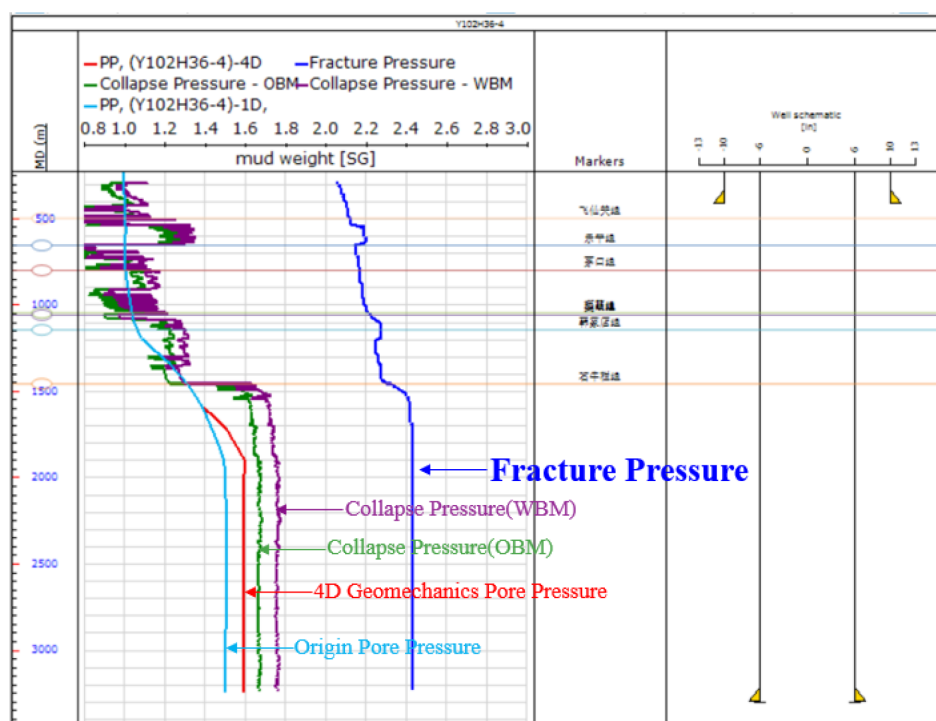


Figure 7—Pressure Prediction Results Before Drilling Yang 102H36-4

Conclusions

In the Zhaotong Sun Anticline area, four sets of conventional source-reservoir-cap rock assemblages are vertically developed. The assemblage directly related to this study consists of: Shiniulan Formation limestone as the reservoir, Longmaxi Formation shale as the source rock, and Hanjiaodian Formation shale as the cap rock. Therefore, the overlying formations above the Feixianguan are almost in a hydrostatic pressure system. Fractures are locally developed in the Maokou and Qixia Formations, and pressure begins to gradually increase in the Hanjiaodian Formation. The pressure in the Longmaxi source rock can reach up to 1.49 g/cm³.

The results show that in the target Longmaxi Formation of the Yang 102H36 well group, the Poisson's ratio ranges from 0.152 to 0.227, the Young's modulus ranges from 36.1 to 65.8 GPa, and the rock strength ranges from 60.7 to 117 MPa. These characteristics indicate a low Poisson's ratio and high Young's modulus, classifying the formation as medium-hard with relatively good drillability.

Comparing the impact of different mud systems on wellbore instability characteristics, empirical data suggests that under the current drilling cycle and mechanical conditions, the estimated density of oil-based mud can be 0.1-0.15 g/cm³ lower than that of water-based mud while still ensuring smooth drilling operations.

Simulations of approximately 100,000 cubic meters of fracturing fluid entering the formation from three horizontal wells in the northern part of the Yang 102H36 platform and the subsequent production characteristics indicate that the "fracturing-production" activities in the northern half have caused an increase in pore pressure in the southern part of the Yang 102H36 platform by 0.6-1.6 MPa. This corresponds to an increase in mud weight of approximately 0.07-0.1 g/cm³.

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